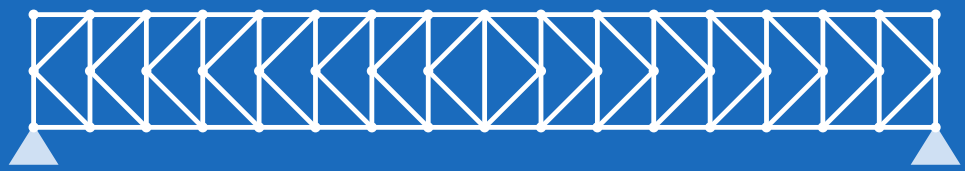
TRUSS REPORT

# K-truss bridge 64 m

Plane truss — analysis and CIRSOC 301 rod checks



Author A. Student · Course Structures I · 13/10/2026

| MAX TENSION | MAX COMPRESSION | DEFLECTION | MOST USED ROD | BUCKLING $\Lambda_1$ | WEIGHT |
|---|---|---|---|---|---|
| +945.0 kN | -945.0 kN | 84.1 mm | 0.73 | 2.34 | 15679 kg |
| rod 8 | rod 24 | L/761 · limit L/300 | rod 8 · normal stress (H.3.4) | the load can grow ×2.3 | carries 900.0 kN · 6× own weight |

## The model

| | |
|---|---|
| Span | 64.00 m |
| Height | 8.00 m |
| Nodes / rods | 50 / 97 |
| Supports | pin at 0, rollerX at 16 |
| Loads | 15 nodal, total 900.0 kN |
| Rod families | 5, with their sections: |
| Bottom chord | HEA 220, F24 (CIRSOC) / A36 |
| Diagonal | SHS 150x150x6, F24 (CIRSOC) / A36 |
| End post | SHS 200x200x8, F24 (CIRSOC) / A36 |
| Top chord | HEB 220, F24 (CIRSOC) / A36 |
| Vertical | SHS 120x120x5, F24 (CIRSOC) / A36 |

## What the analysis found

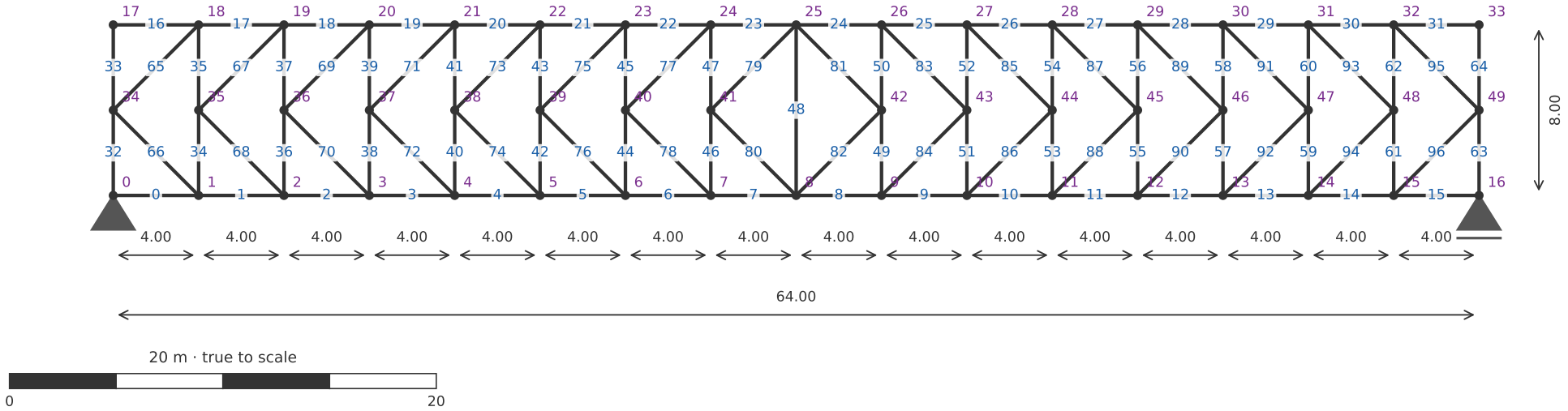
| | |
|---|---|
| Stability | Statically determinate (m + r − 2j = 97 + 3 − 2×50 = 0): statics alone gives every force. |
| Most used rod | rod 8 at 0.73 (normal stress (H.3.4)) |
| Rods over capacity | 0 of 97 checked |
| Buckling | $\lambda_1 = 2.34$ |
| Shear and moment | none: every rod is pinned and loaded only at its nodes, so the axial force is the whole result |

## Contents

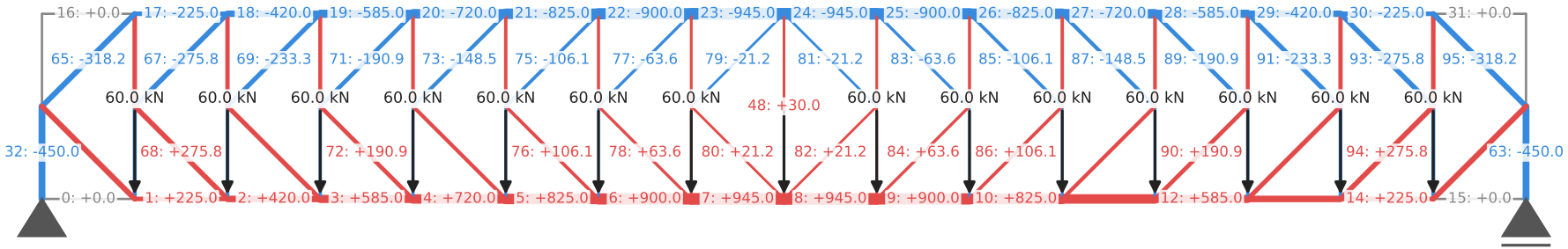

2 Geometry
3 Axial forces
4 Deflection and stress
5 Reactions
6 Utilisation
7 Buckling
8 Rods · 2 sheets
10 Explain rod 8 and 7


Results are an aid to a qualified professional, who must check them. Rod checks to CIRSOC 301 as implemented; first-order analysis; joints and connections are not checked here.

Node numbers in purple, rod numbers in blue. Dimensions in m.



Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 945.0).

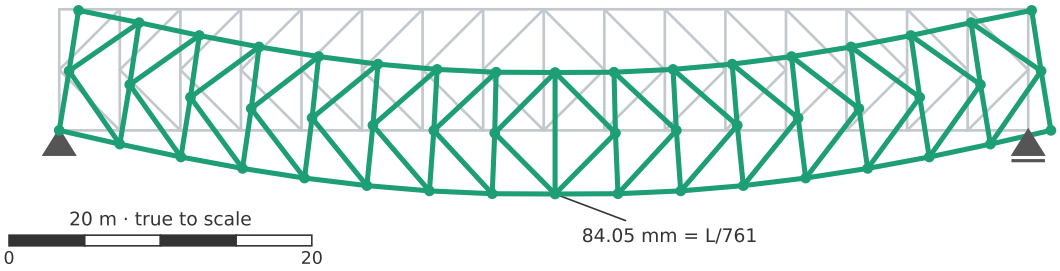



20 m · true to scale

0 20

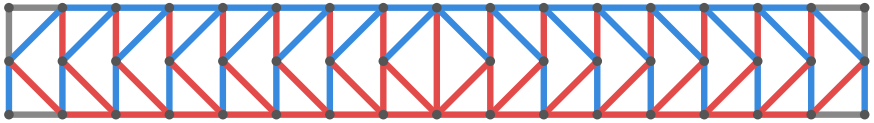
39 of 112 labels left out where they would overlap; every force is in the Rods table.

Elastic line: the deflected shape (green) over the truss (grey), every rigid rod drawn as the curve it bends into. Geometry to scale; displacements magnified × 50. Every rod is pinned, so every member stays straight: there is no curvature, and no contraflexure. Largest vertical deflection 84.05 mm against the limit L/300 = 213.3 mm: OK.

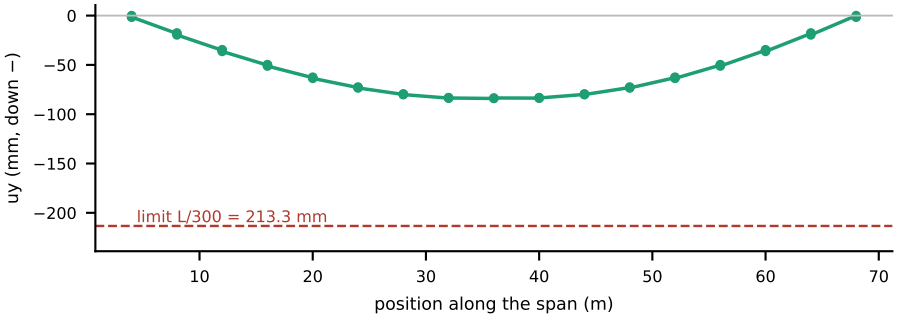



## Fibre stress top and bottom face of each member

red = tension, blue = compression. Every rod is pinned, so both faces of a rod carry the same stress and each rod is one colour: the axial force.

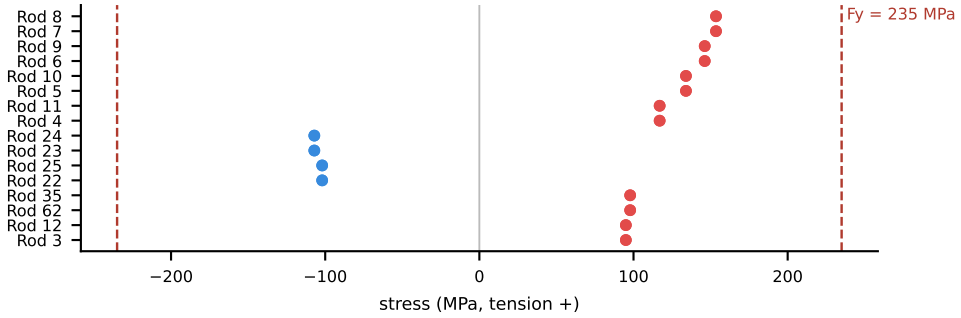


## Deflection along the span vertical movement of every joint



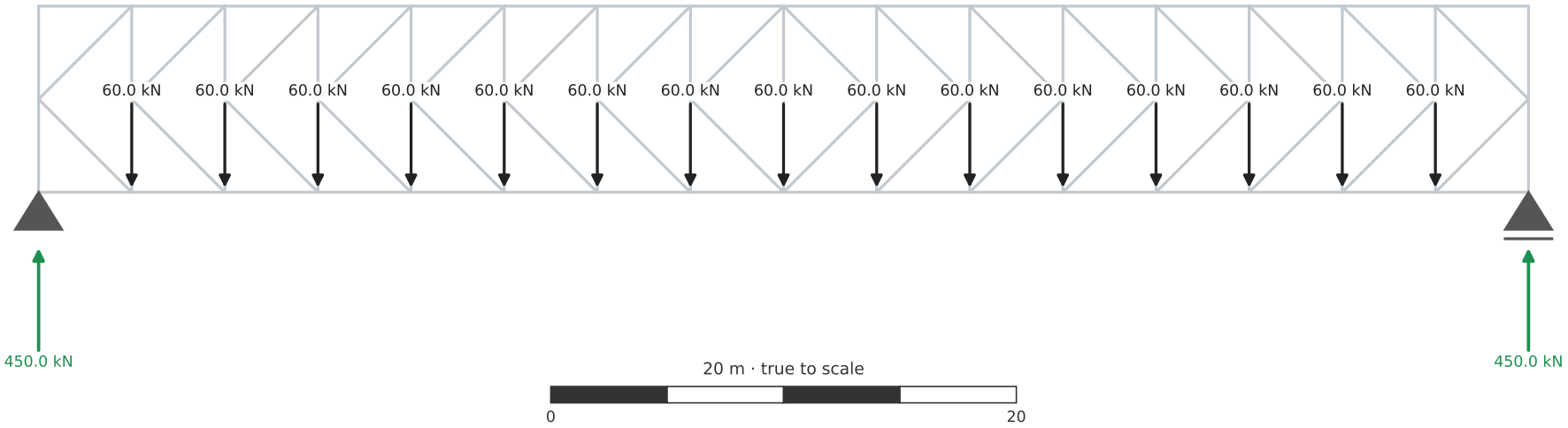

Largest 84.05 mm at joint 8 = L/761. Within the limit.

## Fibre stress values the two faces of each rod, MPa



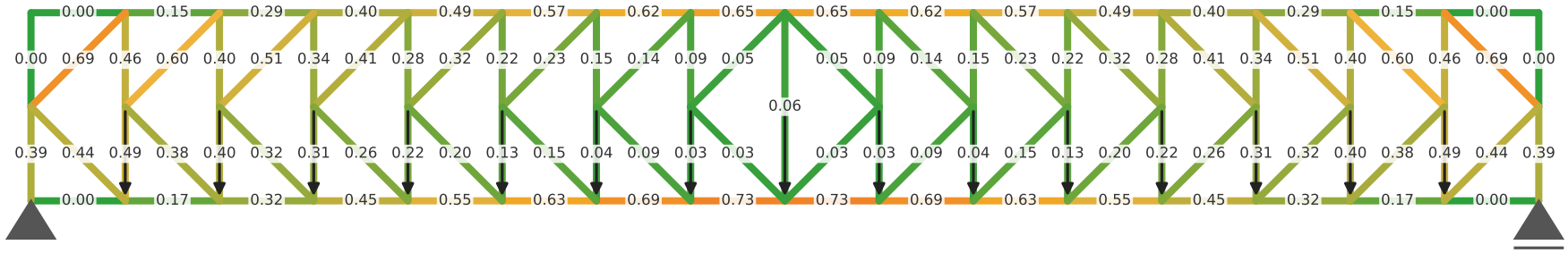

Every rod is pinned, so its two fibres carry the same N/A and coincide: no bending opens the pair.

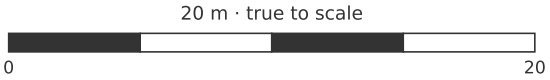
Support reactions (green arrows), as the supports push on the truss; Ry positive upwards. The sums balance the loads.

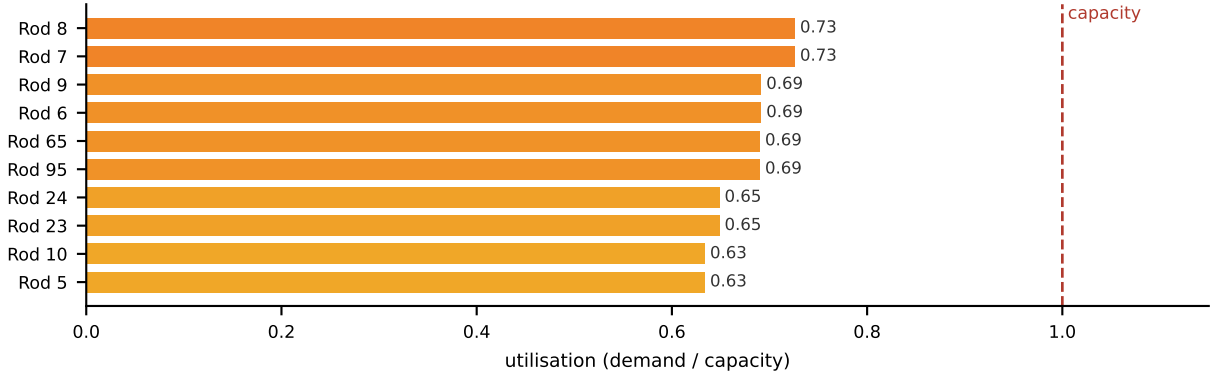


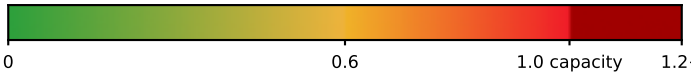

| Node | Type | Rx (kN) | Ry (kN) |
|---|---|---|---|
| 0 | pin | 0.00 | 450.00 |
| 16 | rollerX | 0.00 | 450.00 |
| Σ | | 0.00 | 900.00 |

Utilisation: demand over CIRSOC 301 capacity. Most used rod 8 at 0.73 (normal stress (H.3.4)). Every checked rod is within capacity.



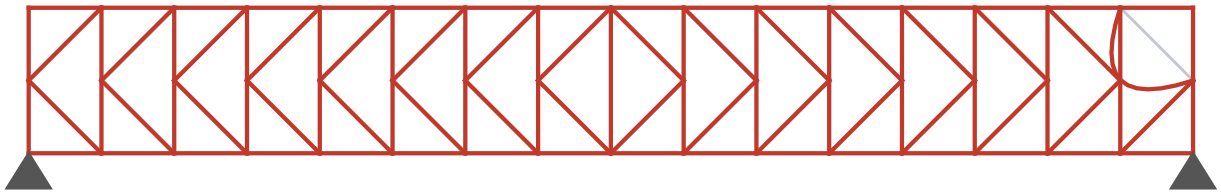




grey = no steel section, not checked

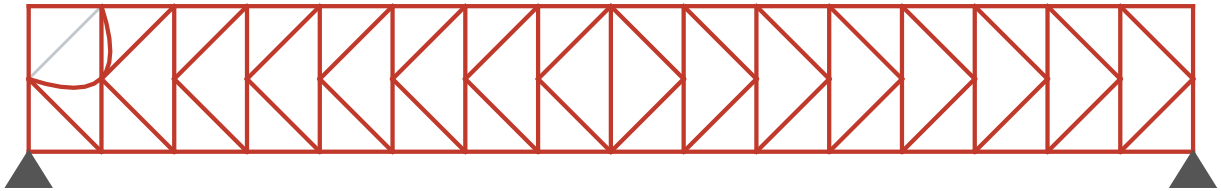
The 10 most used of 97 checked rods; the rest are in the Rods table.

λ is the factor on the load as it stands. $\lambda_1 = 2.34$, so the load can grow 2.3× before it buckles. 2 modes share that λ: that many equal members buckle at the same load, so stiffening one of them buys nothing.

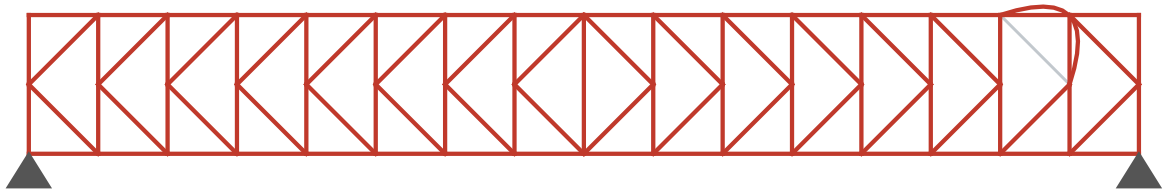
Mode 1: λ = 2.34
rod 95 bows

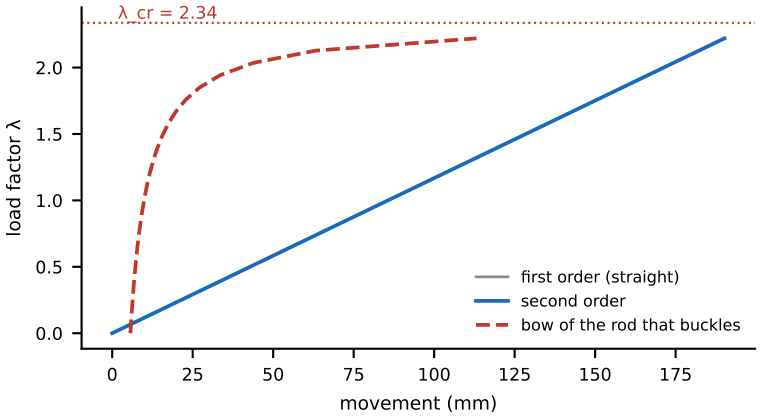


Mode 2: λ = 2.34
rod 65 bows



Mode 3: λ = 2.70
rod 93 bows






The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 5.66 mm in the first mode's shape.

Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

| Rod | Nodes | Family | Section | L (m) | N (kN) | Governing check | Use |
|---|---|---|---|---|---|---|---|
| 0 | 0–1 | Bottom chord | HEA 220 | 4.00 | +0.00 | inelastic buckling (E3-2) | 0.00 OK |
| 1 | 1–2 | Bottom chord | HEA 220 | 4.00 | +225.00 | normal stress (H.3.4) | 0.17 OK |
| 2 | 2–3 | Bottom chord | HEA 220 | 4.00 | +420.00 | normal stress (H.3.4) | 0.32 OK |
| 3 | 3–4 | Bottom chord | HEA 220 | 4.00 | +585.00 | normal stress (H.3.4) | 0.45 OK |
| 4 | 4–5 | Bottom chord | HEA 220 | 4.00 | +720.00 | normal stress (H.3.4) | 0.55 OK |
| 5 | 5–6 | Bottom chord | HEA 220 | 4.00 | +825.00 | normal stress (H.3.4) | 0.63 OK |
| 6 | 6–7 | Bottom chord | HEA 220 | 4.00 | +900.00 | normal stress (H.3.4) | 0.69 OK |
| 7 | 7–8 | Bottom chord | HEA 220 | 4.00 | +945.00 | normal stress (H.3.4) | 0.73 OK |
| 8 | 8–9 | Bottom chord | HEA 220 | 4.00 | +945.00 | normal stress (H.3.4) | 0.73 OK |
| 9 | 9–10 | Bottom chord | HEA 220 | 4.00 | +900.00 | normal stress (H.3.4) | 0.69 OK |
| 10 | 10–11 | Bottom chord | HEA 220 | 4.00 | +825.00 | normal stress (H.3.4) | 0.63 OK |
| 11 | 11–12 | Bottom chord | HEA 220 | 4.00 | +720.00 | normal stress (H.3.4) | 0.55 OK |
| 12 | 12–13 | Bottom chord | HEA 220 | 4.00 | +585.00 | normal stress (H.3.4) | 0.45 OK |
| 13 | 13–14 | Bottom chord | HEA 220 | 4.00 | +420.00 | normal stress (H.3.4) | 0.32 OK |
| 14 | 14–15 | Bottom chord | HEA 220 | 4.00 | +225.00 | normal stress (H.3.4) | 0.17 OK |
| 15 | 15–16 | Bottom chord | HEA 220 | 4.00 | +0.00 | inelastic buckling (E3-2) | 0.00 OK |
| 16 | 17–18 | Top chord | HEB 220 | 4.00 | +0.00 | normal stress (H.3.4) | 0.00 OK |
| 17 | 18–19 | Top chord | HEB 220 | 4.00 | -225.00 | inelastic buckling (E3-2) | 0.15 OK |
| 18 | 19–20 | Top chord | HEB 220 | 4.00 | -420.00 | inelastic buckling (E3-2) | 0.29 OK |
| 19 | 20–21 | Top chord | HEB 220 | 4.00 | -585.00 | inelastic buckling (E3-2) | 0.40 OK |
| 20 | 21–22 | Top chord | HEB 220 | 4.00 | -720.00 | inelastic buckling (E3-2) | 0.49 OK |
| 21 | 22–23 | Top chord | HEB 220 | 4.00 | -825.00 | inelastic buckling (E3-2) | 0.57 OK |
| 22 | 23–24 | Top chord | HEB 220 | 4.00 | -900.00 | inelastic buckling (E3-2) | 0.62 OK |
| 23 | 24–25 | Top chord | HEB 220 | 4.00 | -945.00 | inelastic buckling (E3-2) | 0.65 OK |
| 24 | 25–26 | Top chord | HEB 220 | 4.00 | -945.00 | inelastic buckling (E3-2) | 0.65 OK |
| 25 | 26–27 | Top chord | HEB 220 | 4.00 | -900.00 | inelastic buckling (E3-2) | 0.62 OK |
| 26 | 27–28 | Top chord | HEB 220 | 4.00 | -825.00 | inelastic buckling (E3-2) | 0.57 OK |
| 27 | 28–29 | Top chord | HEB 220 | 4.00 | -720.00 | inelastic buckling (E3-2) | 0.49 OK |
| 28 | 29–30 | Top chord | HEB 220 | 4.00 | -585.00 | inelastic buckling (E3-2) | 0.40 OK |
| 29 | 30–31 | Top chord | HEB 220 | 4.00 | -420.00 | inelastic buckling (E3-2) | 0.29 OK |
| 30 | 31–32 | Top chord | HEB 220 | 4.00 | -225.00 | inelastic buckling (E3-2) | 0.15 OK |
| 31 | 32–33 | Top chord | HEB 220 | 4.00 | +0.00 | inelastic buckling (E3-2) | 0.00 OK |
| 32 | 0–34 | End post | SHS 200x200x8 | 4.00 | -450.00 | inelastic buckling (E3-2) | 0.39 OK |
| 33 | 34–17 | End post | SHS 200x200x8 | 4.00 | +0.00 | normal stress (H.3.4) | 0.00 OK |
| 34 | 1–35 | Vertical | SHS 120x120x5 | 4.00 | -165.00 | inelastic buckling (E3-2) | 0.49 OK |
| 35 | 35–18 | Vertical | SHS 120x120x5 | 4.00 | +225.00 | normal stress (H.3.4) | 0.46 OK |
| 36 | 2–36 | Vertical | SHS 120x120x5 | 4.00 | -135.00 | inelastic buckling (E3-2) | 0.40 OK |
| 37 | 36–19 | Vertical | SHS 120x120x5 | 4.00 | +195.00 | normal stress (H.3.4) | 0.40 OK |
| 38 | 3–37 | Vertical | SHS 120x120x5 | 4.00 | -105.00 | inelastic buckling (E3-2) | 0.31 OK |
| 39 | 37–20 | Vertical | SHS 120x120x5 | 4.00 | +165.00 | normal stress (H.3.4) | 0.34 OK |
| 40 | 4–38 | Vertical | SHS 120x120x5 | 4.00 | -75.00 | inelastic buckling (E3-2) | 0.22 OK |
| 41 | 38–21 | Vertical | SHS 120x120x5 | 4.00 | +135.00 | normal stress (H.3.4) | 0.28 OK |
| 42 | 5–39 | Vertical | SHS 120x120x5 | 4.00 | -45.00 | inelastic buckling (E3-2) | 0.13 OK |
| 43 | 39–22 | Vertical | SHS 120x120x5 | 4.00 | +105.00 | normal stress (H.3.4) | 0.22 OK |
| 44 | 6–40 | Vertical | SHS 120x120x5 | 4.00 | -15.00 | inelastic buckling (E3-2) | 0.04 OK |
| 45 | 40–23 | Vertical | SHS 120x120x5 | 4.00 | +75.00 | normal stress (H.3.4) | 0.15 OK |
| 46 | 7–41 | Vertical | SHS 120x120x5 | 4.00 | +15.00 | normal stress (H.3.4) | 0.03 OK |
| 47 | 41–24 | Vertical | SHS 120x120x5 | 4.00 | +45.00 | normal stress (H.3.4) | 0.09 OK |
| 48 | 8–25 | Vertical | SHS 120x120x5 | 8.00 | +30.00 | normal stress (H.3.4) | 0.06 OK |
| 49 | 9–42 | Vertical | SHS 120x120x5 | 4.00 | +15.00 | normal stress (H.3.4) | 0.03 OK |
| 50 | 42–26 | Vertical | SHS 120x120x5 | 4.00 | +45.00 | normal stress (H.3.4) | 0.09 OK |
| 51 | 10–43 | Vertical | SHS 120x120x5 | 4.00 | -15.00 | inelastic buckling (E3-2) | 0.04 OK |

Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

| Rod | Nodes | Family | Section | L (m) | N (kN) | Governing check | Use |
|---|---|---|---|---|---|---|---|
| 52 | 43–27 | Vertical | SHS 120x120x5 | 4.00 | +75.00 | normal stress (H.3.4) | 0.15 OK |
| 53 | 11–44 | Vertical | SHS 120x120x5 | 4.00 | -45.00 | inelastic buckling (E3-2) | 0.13 OK |
| 54 | 44–28 | Vertical | SHS 120x120x5 | 4.00 | +105.00 | normal stress (H.3.4) | 0.22 OK |
| 55 | 12–45 | Vertical | SHS 120x120x5 | 4.00 | -75.00 | inelastic buckling (E3-2) | 0.22 OK |
| 56 | 45–29 | Vertical | SHS 120x120x5 | 4.00 | +135.00 | normal stress (H.3.4) | 0.28 OK |
| 57 | 13–46 | Vertical | SHS 120x120x5 | 4.00 | -105.00 | inelastic buckling (E3-2) | 0.31 OK |
| 58 | 46–30 | Vertical | SHS 120x120x5 | 4.00 | +165.00 | normal stress (H.3.4) | 0.34 OK |
| 59 | 14–47 | Vertical | SHS 120x120x5 | 4.00 | -135.00 | inelastic buckling (E3-2) | 0.40 OK |
| 60 | 47–31 | Vertical | SHS 120x120x5 | 4.00 | +195.00 | normal stress (H.3.4) | 0.40 OK |
| 61 | 15–48 | Vertical | SHS 120x120x5 | 4.00 | -165.00 | inelastic buckling (E3-2) | 0.49 OK |
| 62 | 48–32 | Vertical | SHS 120x120x5 | 4.00 | +225.00 | normal stress (H.3.4) | 0.46 OK |
| 63 | 16–49 | End post | SHS 200x200x8 | 4.00 | -450.00 | inelastic buckling (E3-2) | 0.39 OK |
| 64 | 49–33 | End post | SHS 200x200x8 | 4.00 | +0.00 | normal stress (H.3.4) | 0.00 OK |
| 65 | 34–18 | Diagonal | SHS 150x150x6 | 5.66 | -318.20 | inelastic buckling (E3-2) | 0.69 OK |
| 66 | 34–1 | Diagonal | SHS 150x150x6 | 5.66 | +318.20 | normal stress (H.3.4) | 0.44 OK |
| 67 | 35–19 | Diagonal | SHS 150x150x6 | 5.66 | -275.77 | inelastic buckling (E3-2) | 0.60 OK |
| 68 | 35–2 | Diagonal | SHS 150x150x6 | 5.66 | +275.77 | normal stress (H.3.4) | 0.38 OK |
| 69 | 36–20 | Diagonal | SHS 150x150x6 | 5.66 | -233.35 | inelastic buckling (E3-2) | 0.51 OK |
| 70 | 36–3 | Diagonal | SHS 150x150x6 | 5.66 | +233.35 | normal stress (H.3.4) | 0.32 OK |
| 71 | 37–21 | Diagonal | SHS 150x150x6 | 5.66 | -190.92 | inelastic buckling (E3-2) | 0.41 OK |
| 72 | 37–4 | Diagonal | SHS 150x150x6 | 5.66 | +190.92 | normal stress (H.3.4) | 0.26 OK |
| 73 | 38–22 | Diagonal | SHS 150x150x6 | 5.66 | -148.49 | inelastic buckling (E3-2) | 0.32 OK |
| 74 | 38–5 | Diagonal | SHS 150x150x6 | 5.66 | +148.49 | normal stress (H.3.4) | 0.20 OK |
| 75 | 39–23 | Diagonal | SHS 150x150x6 | 5.66 | -106.07 | inelastic buckling (E3-2) | 0.23 OK |
| 76 | 39–6 | Diagonal | SHS 150x150x6 | 5.66 | +106.07 | normal stress (H.3.4) | 0.15 OK |
| 77 | 40–24 | Diagonal | SHS 150x150x6 | 5.66 | -63.64 | inelastic buckling (E3-2) | 0.14 OK |
| 78 | 40–7 | Diagonal | SHS 150x150x6 | 5.66 | +63.64 | normal stress (H.3.4) | 0.09 OK |
| 79 | 41–25 | Diagonal | SHS 150x150x6 | 5.66 | -21.21 | inelastic buckling (E3-2) | 0.05 OK |
| 80 | 41–8 | Diagonal | SHS 150x150x6 | 5.66 | +21.21 | normal stress (H.3.4) | 0.03 OK |
| 81 | 42–25 | Diagonal | SHS 150x150x6 | 5.66 | -21.21 | inelastic buckling (E3-2) | 0.05 OK |
| 82 | 42–8 | Diagonal | SHS 150x150x6 | 5.66 | +21.21 | normal stress (H.3.4) | 0.03 OK |
| 83 | 43–26 | Diagonal | SHS 150x150x6 | 5.66 | -63.64 | inelastic buckling (E3-2) | 0.14 OK |
| 84 | 43–9 | Diagonal | SHS 150x150x6 | 5.66 | +63.64 | normal stress (H.3.4) | 0.09 OK |
| 85 | 44–27 | Diagonal | SHS 150x150x6 | 5.66 | -106.07 | inelastic buckling (E3-2) | 0.23 OK |
| 86 | 44–10 | Diagonal | SHS 150x150x6 | 5.66 | +106.07 | normal stress (H.3.4) | 0.15 OK |
| 87 | 45–28 | Diagonal | SHS 150x150x6 | 5.66 | -148.49 | inelastic buckling (E3-2) | 0.32 OK |
| 88 | 45–11 | Diagonal | SHS 150x150x6 | 5.66 | +148.49 | normal stress (H.3.4) | 0.20 OK |
| 89 | 46–29 | Diagonal | SHS 150x150x6 | 5.66 | -190.92 | inelastic buckling (E3-2) | 0.41 OK |
| 90 | 46–12 | Diagonal | SHS 150x150x6 | 5.66 | +190.92 | normal stress (H.3.4) | 0.26 OK |
| 91 | 47–30 | Diagonal | SHS 150x150x6 | 5.66 | -233.35 | inelastic buckling (E3-2) | 0.51 OK |
| 92 | 47–13 | Diagonal | SHS 150x150x6 | 5.66 | +233.35 | normal stress (H.3.4) | 0.32 OK |
| 93 | 48–31 | Diagonal | SHS 150x150x6 | 5.66 | -275.77 | inelastic buckling (E3-2) | 0.60 OK |
| 94 | 48–14 | Diagonal | SHS 150x150x6 | 5.66 | +275.77 | normal stress (H.3.4) | 0.38 OK |
| 95 | 49–32 | Diagonal | SHS 150x150x6 | 5.66 | -318.20 | inelastic buckling (E3-2) | 0.69 OK |
| 96 | 49–15 | Diagonal | SHS 150x150x6 | 5.66 | +318.20 | normal stress (H.3.4) | 0.44 OK |



## Rod 8

utilisation 0.73

normal stress (H.3.4)

```
Rod 8
=====
Steel, CIRSOC 301-2018 -- A = 6,156 mm², Fy = 235 MPa, L = 4.000 m

 1. Axial stress
      σ = N / A
      = 945,000 N / 6,156 mm²
      = 153.51 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 153.51 / 211.50
      = 0.726

 4. Governs
      the largest ratio
      = normal stress (H.3.4)
      = 0.726

utilisation 0.726 -- OK
```



## Rod 7

utilisation 0.73

normal stress (H.3.4)

```
Rod 7
=====
Steel, CIRSOC 301-2018 -- A = 6,156 mm², Fy = 235 MPa, L = 4.000 m

 1. Axial stress
      σ = N / A
      = 945,000 N / 6,156 mm²
      = 153.51 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 153.51 / 211.50
      = 0.726

 4. Governs
      the largest ratio
      = normal stress (H.3.4)
      = 0.726

utilisation 0.726 -- OK
```